23ISRLTSTT2P1

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Problem

For positive integers n, let $f_2(n)$ denote the number of divisors of n which are perfect squares, and $f_3(n)$ denotes the number of positive divisors which are perfect cubes. Prove that for each positive integer k there exists a positive integer n for which $\dfrac{f_2(n)}{f_3(n)} = k$.

Solution. Let $n = \prod_{i=1}^{m} p_i^{6\beta_i + j_i}$ for non negative integers β_i, j_i $\forall 1 \le i \le m$.

We can conclude that $f_2(n) = \prod_{i=1}^{m} \left(3\beta_i + \left\lfloor \frac{j_i}{2} \right\rfloor + 1\right)$ and $f_3(n) = \prod_{i=1}^{m} \left(2\beta_i + \left\lfloor \frac{j_i}{3} \right\rfloor + 1\right)$.

For any β, as j varies from 0 to 5, $\left(3\beta + \left\lfloor \frac{j}{2} \right\rfloor + 1, 2\beta + \left\lfloor \frac{j}{3} \right\rfloor + 1\right)$ takes the values of the form $(3x+1, 2x+1), (3x+2, 2x+1), (3x+2, 2x+2), (3x+3, 2x+2)$ for non negative integers x.

So we can say that we have to represent k as a product of rationals of the form $\frac{3x+1}{2x+1}, \frac{3x+2}{2x+1}, \frac{3x+2}{2x+2}, \frac{3x+3}{2x+2}$.

We claim that if a and b can be represented, then ab can be represented as well. First of all, we take a note that the prime divisors do not impact f_2 or f_3, it is their exponent. As a result, if we switch one of the prime divisors of a number with another prime, the values of f_2, f_3 will not change. Say $n_1 = \prod_{i=1}^{a_1} p_i^{6\alpha_i + g_i}$ works for a and $n_2 = \prod_{i=1}^{b_1} q_i^{6\beta_i + h_i}$ works for b for primes p_is and q_is. We set p_is and q_is such that no two of them are equal. As a result

$$\begin{aligned}
\frac{f_2(n_1 n_2)}{f_3(n_1 n_2)} &= \frac{\left(\prod_{i=1}^{a_1} \left(3\alpha_i + \left\lfloor \frac{g_i}{2} \right\rfloor + 1\right)\right)\left(\prod_{i=1}^{b_1} \left(3\beta_i + \left\lfloor \frac{h_i}{2} \right\rfloor + 1\right)\right)}{\left(\prod_{i=1}^{a_1} \left(2\alpha_i + \left\lfloor \frac{g_i}{3} \right\rfloor + 1\right)\right)\left(\prod_{i=1}^{b_1} \left(2\beta_i + \left\lfloor \frac{h_i}{3} \right\rfloor + 1\right)\right)} \\
&= \frac{f_2(n_1)}{f_3(n_1)} \cdot \frac{f_2(n_2)}{f_3(n_2)} \\
&= ab
\end{aligned}$$

We claim that we can represent all primes $p \geq 5$ of this form using induction.
For base case, $(k, n) = (1, 1), (2, 2^{11} \cdot 3^6), (3, 2^{11} \cdot 3^{11} \cdot 5^6), (4, 2^{11} \cdot 3^{11} \cdot 5^6 \cdot 7^6)$ work.

For the inductive step, assume that this works till $1, 2, \ldots, q$ for some prime q. Consider the prime, q' that appears just after q.
We can conclude that we can represent all numbers till $q' - 1$ since they contain only prime divisors that are smaller than or equal to q, so by the above proven lemma, they can be broken into their prime factorisation and the lemma can be used repeatedly for each prime at a time.
Now say q' is of the form $6f + 1$. Consider n_0 to be the number such that $\dfrac{f_2(n_0)}{f_3(n_0)} = 4f + 1$.
As a result, $n_0 \cdot y^{12f}$ suffices (where y is a prime picked such that $y \nmid n_0$).
If q' is of the form $6f - 1$ then consider n' to be the number such that $\dfrac{f_2(n')}{f_3(n')} = 4f + 1$.
As a result, $n' \cdot z^{12f+2}$ suffices (where z is a prime picked such that $z \nmid n'$).
Since we can represent q' as well, induction proves that we can do this for all primes.
Since we can do this for all primes, for any k, we may represent it in form of its primes and use the multiplicativity claim repeatedly.